

Load prediction method for heat and electricity demand in buildings for the purpose of planning for mixed energy distribution systems

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Abstract

Energy planning for mixed energy distribution systems is important to increase the flexibility in the regional and national energy systems. Expected maximum loads, load profiles and yearly energy demands, all divided into heat and electricity purposes, are important input parameters to plan for the most economical, technical and environmental optimal energy distribution system for a planning area. First, this article presents a load prediction method which estimates heat and electricity load profiles for various building categories. The method is based on statistical analyses of hourly simultaneous measured district heat and electricity consumption in several buildings, as well as background information of the measured buildings. The heat load model is based on regression analyses, whereas the electricity load model is based on various statistical distributions. Second, a method for load aggregation based on the building categories' load profiles is presented to estimate the maximum load demands, yearly load profiles, load duration profiles and yearly energy demands, all divided into heat and electricity purposes, for a planning area.

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1. Introduction

Energy planning for mixed energy distribution systems, i.e. energy distribution systems for a specified planning area incorporating more than one energy carrier, is important to increase the flexibility in the regional and national energy systems. Simultaneous heat and electricity load demands are important input parameters when planning for the most economical, technical and environmental optimal energy distribution system for a specified area.

The objective of this article is to present a method developed for load modeling of buildings in mixed energy distribution systems, i.e. estimation of load profiles and yearly energy demand for various building categories divided into heat and electricity purposes. Heat load includes the end-uses space heating, ventilation heating and hot service water. Electricity load includes the end-uses lighting, pumps, fans and electrical

appliances. Cooling as an end-use has not been thoroughly investigated due to the limited amount of electricity measurements including this end-use. However, the method can also be applied when such measurements become available. The load prediction method can estimate expected maximum loads, load profiles and yearly energy demands, all divided into heat and electricity purposes for a planning area.

There exist many methods for estimation of thermal load and energy consumption in buildings. The different methods can broadly be divided in three groups: statistical usually regression analysis, intelligent computer systems, and energy simulation. Regression analysis is mainly based on long-term measurements of load and weather data. Intelligent computer systems in addition require registration of the behavior of the occupants. Energy simulation programs require detailed information about the building, energy systems and behavior of the occupants in addition to a statistical representation of weather data. A complementary literature survey of the different methods can be found in: Use of different methods for thermal load and energy estimations in building including meteorological and sociological input parameters [1].

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2. Load prediction method

Hourly simultaneous district heat and electricity in several buildings within the building categories single family houses (SH) and apartment blocks (AB), office buildings (OB), educational buildings (EB), hotels and restaurants (HR) and hospital buildings (HB) have been collected in Trondheim and Bergen, Norway. Statistical analyses of the district heat and electricity measurements were performed to develop a method for estimating load profiles for the various building categories.

The division of building categories is based on the Directive on the energy performance of buildings [2], as well as the concept of archetypes [3]. The latter is a division of buildings

based on criteria such as technology, market, regulation and design [4]. Background information concerning the buildings analyzed as well as climatic parameters from the locales were collected.

Fig. 1 shows a flow chart of the method developed for estimating relative heat and electricity load profiles. The heat and electricity load model will be presented in the next sections, as well as the method for load aggregation.

2.1. Heat load model

The heat load model is based on piece-wise linear regression analyses of daily mean temperature versus hourly measured

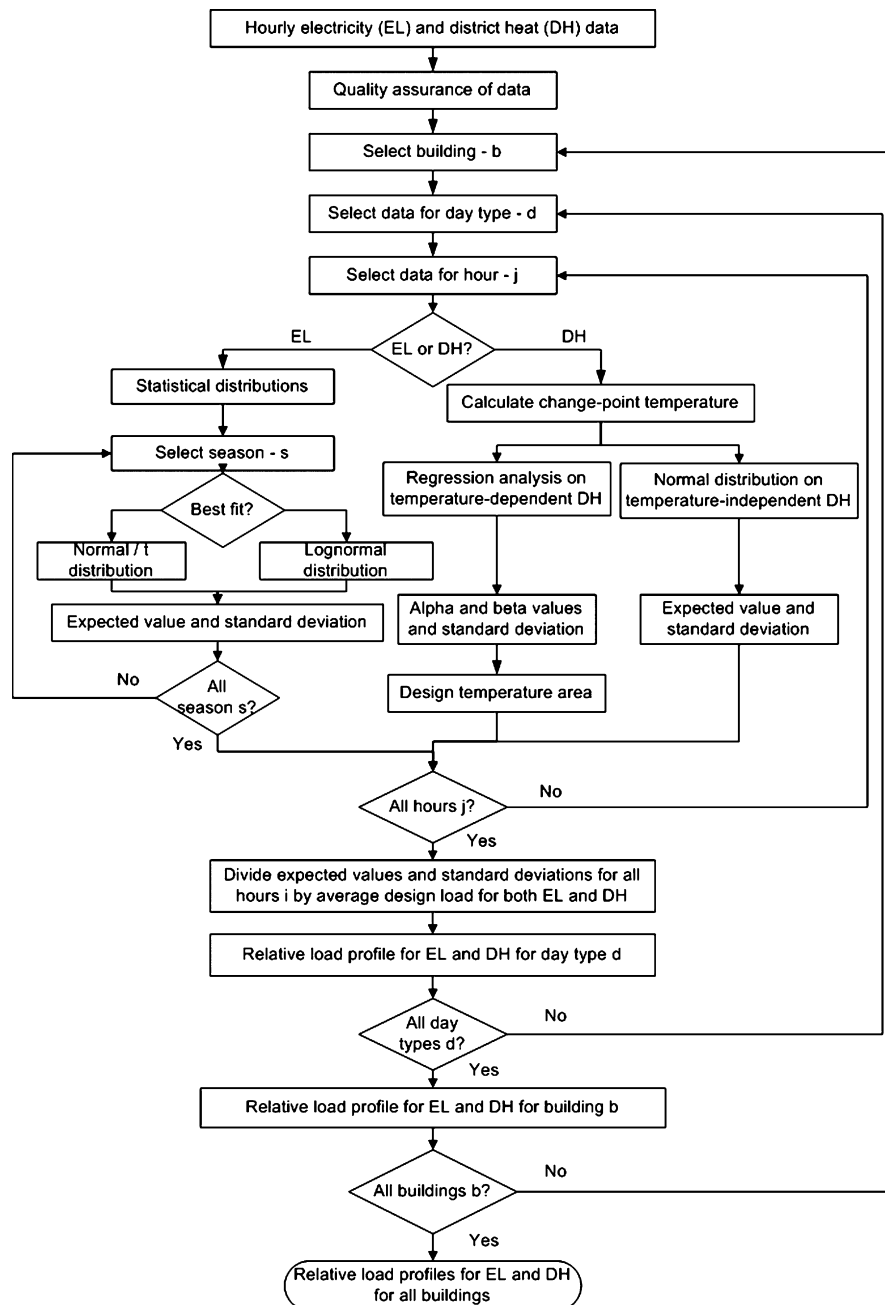


Fig. 1. Flow chart showing the method for the estimation of relative heat and electricity load profiles for individual buildings [4].

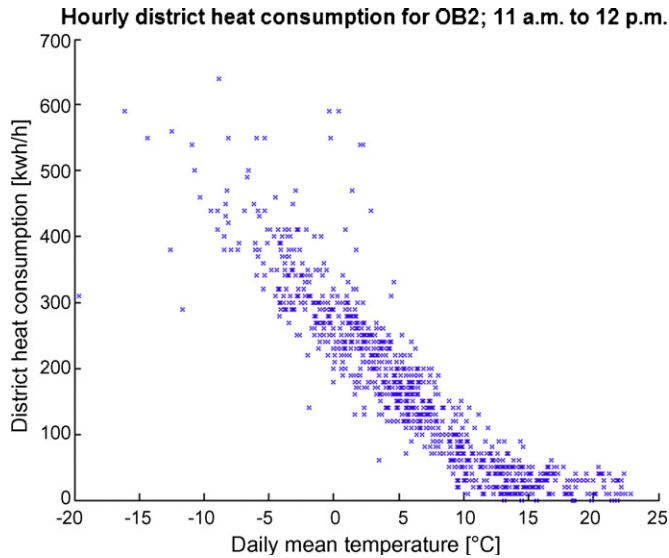


Fig. 2. Scatter plot of daily mean temperature vs. hourly district heat consumption for OB2 in Trondheim for weekdays hour 12 [4].

district heat consumption for every hour of the day and day type. The day types are divided into weekdays (WD); Mondays through Fridays, and weekends/holidays (WE); mainly Saturdays and Sundays. Fig. 2 shows a scatter plot of the daily mean temperature versus the hourly heat consumption for an office building, OB2, in Trondheim hour 12, i.e. from 11 a.m. to 12 p.m., weekdays. The latter figure shows the difference between the temperature-dependent and temperature-independent district heat consumption. As a consequence, the two different parts have to be analyzed separately. The change-point temperature which separates the temperature-dependent and temperature-independent district heat consumption has to be found.

A linear regression equation is expressed as

$$Y_i = \alpha + \beta x_i + e_i \quad (1)$$

where x_i is the independent regressor variable and Y_i is the dependent random variable. e_i is the called the residual and describes the error in the fit of the model [6].

The change-point temperature is found by assuming a linear correlation between the daily mean temperature and the hourly district heat consumption. The regression coefficients α and β are calculated for temperature steps of 0.1 °C, starting at an initial change-point temperature of 17 °C and stepping down to 0 °C. The regression coefficients α and β are estimated based on the least square method [6]. The change-point temperature is found in the range where the β -values fluctuate least, i.e. an approximately constant β -value indicates that the influence of the temperature-independent heat consumption is neglectable.

The mathematical procedure developed to find the change-point temperature for a given building at a given hour is based on α - and β -values in the following way [5]:

1. Calculate the temperature range of $\Delta\theta = 5$ °C in which the total sum of squares of the α -values is smallest. The α - and β -values are calculated in advance for change-point tempera-

tures from 17 °C and down to 0 °C with a temperature step of 0.1 °C.

2. Calculate the temperature band of $\Delta\theta = 1$ °C where the minimum total sum of squares of the β -values occur within the temperature range of $\Delta\theta = 5$ °C found in number 1. The temperature band slide from the highest temperature in the temperature range to the lowest with a temperature step of 0.1 °C.
3. The temperature-dependent season is then found within the temperature band given in number 2. The change-point temperature is defined as the average temperature within the temperature band.

More information concerning the mathematical procedure and the verification of the heat load model can be found in [5].

2.1.1. Temperature-dependent heat load model

The heat load demand for a given hour j and day type d is shown in the following equation:

$$\Phi_{HL,j,d} = \alpha_{j,d} + \beta_{j,d}\theta + e_{j,d} \quad (2)$$

where $e_{j,d}$ is the residuals; $\alpha_{j,d}$, $\beta_{j,d}$ the regression coefficients; $\Phi_{HL,j,d}$ the heat load demand for a given hour j and day type d ; θ the daily mean temperature; d the day type; weekday (WD) or weekend/holiday (WE); j the 1, 2, 3, ..., 24 where 1 = 12 a.m., ..., 24 = 11 p.m. to 12 a.m.

Linear regression analysis is performed on every hour of the day for each day type for the temperature-dependent district heat consumption. The latter data included all the hourly district heat measurements below the estimated change-point temperature.

The design heat load profile is found by inserting the design temperature into Eq. (2) for every hour and day type. The design temperature in Norway, θ_{dt} , is defined as the average outdoor temperature during the three coldest successive days in a 30-year period; the period from 1961 to 1990 apply presently.

2.1.2. Temperature-independent heat load model

The temperature-independent heat load model is based on the hourly district heat consumption during the temperature-independent season, i.e. hourly district heat measurements above the change-point temperature. This mainly represents hot service water consumption. The model is based on the assumption that the heat load above the change-point temperature is independent of outdoor temperature. Consequently, the temperature-independent heat load is investigated in relation to various probability distributions, and the expected values and standard deviations for every hour and day type for all buildings analyzed are calculated. The normal distribution and the Student's t -distribution showed the best fit.

2.1.3. Relative values for heat load profiles

It was important to produce relative load profiles [7] for the reason of comparison, either by archetype or building category. An archetype is a division of buildings based on other criteria than building category, and different archetype classifications

may be technology, market, user behavior, regulations and design, planning and construction processes [8]. The main findings of archetype division in relation to heat load profiles are the operation of the ventilation systems and the construction year or major retrofitting of the buildings.

The average daily design load, $\bar{\Phi}_{HL,d}$, is calculated according to the following equation:

$$\bar{\Phi}_{HL,d} = \frac{1}{24} \sum_{j=1}^{24} \Phi_{HL,j,d}(\theta_{dt}) = \frac{\sum_{j=1}^{24} (\alpha_{j,d} + \beta_{j,d} \theta_{dt})}{24} \quad (3)$$

The design conditions for heat load always occurred during weekdays for the buildings analyzed.

Relative design heat load profiles for all buildings within a given building category are derived for the reason of comparison and generalization of the heat load profiles. The relative design load profiles are found by dividing the design heat load for each hour and day type by the average daily design load. The relative heat load profiles for the temperature-independent heat consumption are also found by dividing the expected values for each hour and day type by the average design load.

The relative design heat load profile weekdays for an office building, OB7, is shown in Fig. 3 along with the standard deviation bounds. The y-axis is given in P.U.; meaning per unit or relative to one. This representation allowed for comparisons despite the difference in design heat load.

2.1.4. Generalization of heat load profiles

The buildings have to be sorted into different archetypes based on building type and regulation regime to generalize the heat load profiles. The relative expected value for each archetype is calculated based on each building's relative expected value after the buildings analyzed have been divided into different archetypes and building categories.

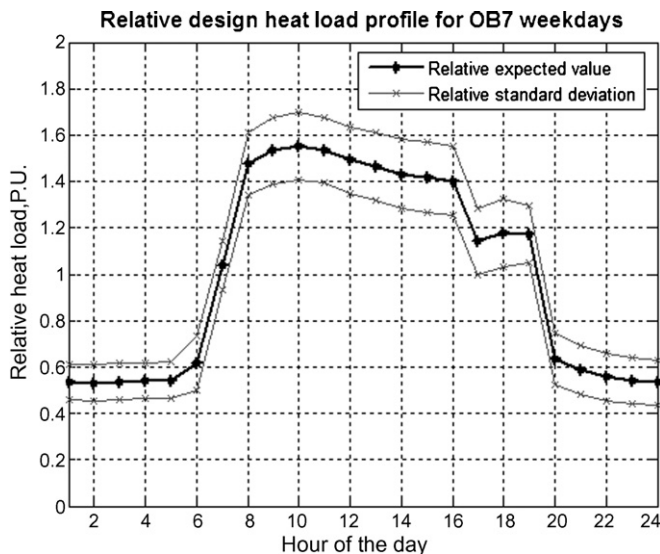


Fig. 3. Relative design heat load profile for OB7 in Trondheim for weekdays, including relative standard deviation bounds [4].

If $X_1, X_2, \dots, X_n$ are independent variables from the same probability distribution with mean value, μ , and standard deviation, σ , then the central limit theorem states that [6]:

$$\bar{X} = \frac{1}{n} (X_1 + X_2 + \dots + X_n) \quad (4)$$

is approximately $\text{Normal}(\mu, \sigma/\sqrt{n})$. The sample size, N , within each archetype/building category should be larger than 30 to use the normal distribution when generalizing the heat load profiles.

Eq. (4) implies that the expected relative heat load for every hour j and day type d for a given archetype can be expressed as

$$\bar{\Phi}_{HL,j,d} = \frac{1}{n} \sum_{n=1}^N \Phi_{HL,j,d,n} \quad (5)$$

where n is the number of buildings within the selected archetype or building category; $\bar{\Phi}_{HL,j,d}$ the relative expected heat load for an archetype or building category hour j and day type d .

The regression coefficients α and β are inserted into Eq. (5) for every building within the archetype or building category at a given hour and day type:

$$\bar{\alpha}_{j,d} + \bar{\beta}_{j,d} \theta = \frac{1}{n} \sum_{n=1}^N \alpha_{j,d,n} + \frac{\theta}{n} \sum_{n=1}^N \beta_{j,d,n} \quad (6)$$

2.2. Electricity load model

The electricity load model is based on probability distribution analyses for every hour of the day and day type. The t -test is applied to the hourly electricity consumption in relation to daily mean temperature to reveal temperature-dependencies. All buildings which show temperature-dependent electricity consumption are omitted from the analyses based on the criteria of end-use division, i.e. electricity as an energy carrier should only supply lighting, pumps, fans and electrical appliances.

2.2.1. Probability distributions

To analyze the electricity load demand in buildings supplied by both district heating and electricity, various probability distributions are applied. The electricity load model is based on continuous probability distributions. The hourly electricity consumption data for each hour and day type are mainly examined in relation to normal, lognormal and Student's t -distributions, but also other probability distributions have been tested [5]. See [6] for more information about statistical analyses in general and probability distributions in particular.

Probability plots are used to analyze the goodness of fit for the various distributions. The expected value and the standard deviation for each hour and day type for every building are calculated. A probability plot of an office building, OB2, in Trondheim for weekdays hour 12 is shown in Fig. 4. The Student's t -distribution (or T -scale distribution) gives a good fit. The normal distribution does not fit as well and the lognormal distribution does not fit for this particular high load hour.

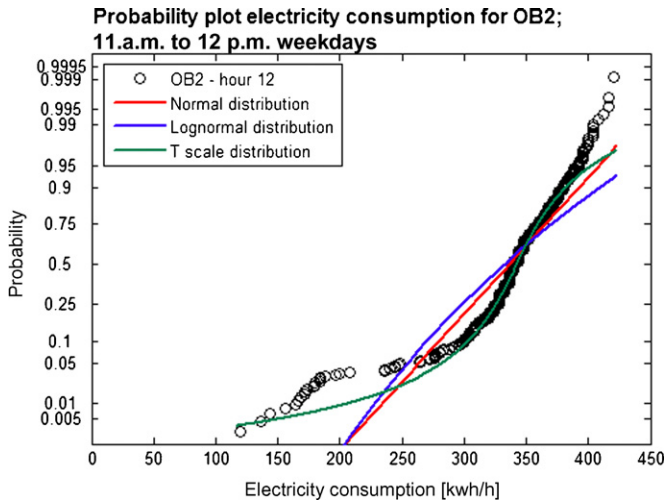


Fig. 4. Probability plot of hourly electricity consumption for OB2 weekdays between 11 a.m. and 12 p.m. The goodness of fit is shown for normal, lognormal and Student's *t*-distribution [4].

The analyses of the electricity consumption revealed that there are some seasonal variations which cannot be related to outdoor temperature alone, see [5] for more information. As a result, the electricity consumption is investigated in relation to hour of the day and day types, as well as various seasons. Monthly analyses of the electricity load led to the seasonal division of winter (December, January and February), summer (June, July and August) and finally spring/fall; including the remaining months.

2.2.2. Relative values for electricity load profiles

The electricity load profiles are also derived on relative form for the reason of comparison. The design conditions for electricity load profiles occurred during weekdays for the winter season for all buildings analyzed. The average daily design load for electricity, $\bar{\Phi}_{HL,d,s}$, is based on the average expected value for the weekday winter season load profile. The relative electricity load profiles for each season are found by dividing the expected electricity load for each hour and day type by the average design electricity load.

2.2.3. Generalization of electricity load profiles

Generalized electricity load profiles have also been calculated for every archetype or building category for each day type and season. Eq. (7) expresses the expected relative electricity load for every hour *j*, day type *d* and season *s* for a given archetype/building category:

$$\bar{\Phi}_{EL,j,d,s} = \frac{1}{n} \sum_{n=1}^N \Phi_{EL,j,d,s,n} \quad (7)$$

$\bar{\Phi}_{EL,j,d,s}$ is the relative expected electricity load for an archetype or building category hour *j*, day type *d* and season *s*.

The relative electricity load profiles for all office buildings weekdays, including the relative standard deviations for the winter season, are shown in Fig. 5. The generalized winter electricity load profile for the office building category is plotted

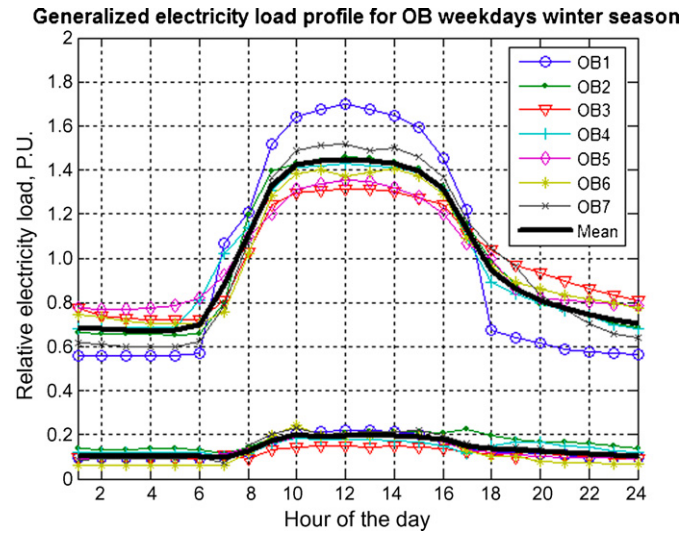


Fig. 5. Generalized electricity load profile, including standard deviation for weekdays during the winter season for the office building category [4].

along with the standard deviation; see the black bold lines in Fig. 5.

2.3. Load aggregation method

The procedure for using the generalized relative load profiles to estimate the design load profiles, the yearly load profiles and the annual energy demands for a specified planning area, all divided into heat and electricity, will be presented in the following sections.

The energy distribution systems include both electrical cables/wires and pipelines for district heating supply. The maximum load losses and annual energy losses from the distribution systems have to be incorporated when aggregating the heat and electricity load from the building level to the energy production unit. The coincidence factors for heat and electricity loads are also discussed in relation to aggregated load profiles, whereas the energy production efficiencies are considered outside the system boundaries, and consequently, not discussed.

2.3.1. Background

If $X_1, X_2, \dots, X_n$ are independent variables from the same distribution with means $\mu_1, \mu_2, \dots, \mu_n$ and variance $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, respectively, then the sum of the independent variables are calculated according to the following equation [6]:

$$Y = X_1 + X_2 + \dots + X_n \quad (8)$$

The sum of the variables has an approximately normal distribution with $\text{Normal}(n\mu, \sqrt{n}\sigma)$.

A bottom-up approach is applied for the aggregation of individual building load profiles to derive the heat and electricity load profiles for a specified planning area supplied by mixed energy distribution systems. The load aggregating method is based on the sum of normal distributions. It is assumed that the load profiles developed for heat and electricity

Table 1

Specific heat and electricity load for the various archetypes/building categories analyzed based on design temperature of $-19\text{ }^{\circ}\text{C}$ and winter season load profile [4]

Archetype/building category	Specific load (W/m^2)			
	HL WD	HL WE	EL WD	EL WE
Single family houses and apartment blocks	46.0	44.7	10.5	10.3
Office buildings	55.6	44.5	23.8	13.0
Educational buildings: before 1997	61.3	34.0	19.6	6.3
Educational buildings: after 1997	81.3	29.2		
Retirement homes (hospital buildings)	64.0	59.4	23.1	20.2
Hotels with restaurants	42.6	41.1	16.3	15.9

for different archetypes/building categories are independent and normally distributed.

2.3.2. Aggregated design load

The distribution intervals for a normally distributed variable X can be written as follows [8]:

1. It is a $100(1 - \alpha)\%$ certainty that the X -value occurs within the interval $\mu \pm (z_{\alpha/2}\sigma)$.
2. It is a $100(1 - \alpha)\%$ certainty that the X -value will be less than $\mu + (z_{\alpha}\sigma)$.
3. It is a $100(1 - \alpha)\%$ certainty that the X -value will be greater than $\mu - (z_{\alpha}\sigma)$.

The value z_{α} is called the α -quantile. The latter values are tabulated for various distributions such as the normal and Student's t -distribution. The α -values here represent the level of significance and must not be mistaken for the regression coefficient. The standard deviations are unknown and have to be calculated based on the hourly measured load data. As a consequence, the α -quantiles from the Student's t -distribution, t_{α} , are used.

The distribution interval expressed in number 2 in the list above is applied for aggregated design load. The design conditions for maximum heat and electricity load profiles are dependent on the accuracy level, but in general terms the maximum load may be expressed as Eq. (9) [5]. This approach has also been applied by [7] and by [9] for all electric buildings:

$$\Phi_{\text{MaxLoad}} = \mu_{\text{MaxLoad}} + t_{\alpha}\sigma_{\text{MaxLoad}} \quad (9)$$

μ_{MaxLoad} $\Phi_{\text{HL,Max}}$ or $\Phi_{\text{EL,Max}}$ for heat and electricity, respectively.

A probability of the maximum load occurring below a given value with a 95% likelihood is a good estimate for load modeling of heat and electricity demand for mixed energy distribution systems. Consequently, the 95% α -quantile is chosen for design conditions [5].

2.3.3. Indicators

Specific heat and electricity load indicators, in $[\text{W}/\text{m}^2]$, as well as specific heat and electricity consumption indicators, in $[\text{kWh}/(\text{m}^2 \text{ year})]$, are applied to convert the relative generalized load profiles into real design load profiles and yearly load profiles for a selected building.

Specific heat and electricity loads are calculated for every archetype and building category based on the load prediction method. The real design heat and electricity loads for each building analyzed are calculated, as well as the corresponding standard deviations. The purpose of the specific loads is to restore the generalized design load profiles [5].

Table 1 shows the specific heat and electricity loads for all archetypes/building categories analyzed based on the design temperature in Trondheim of $-19\text{ }^{\circ}\text{C}$ and the winter season load profile.

Energy consumption indicators (ECI) divided into heat (HCI) and electricity (ELCI) purposes are calculated for every archetype/building category analyzed. The temperature-dependent part of the HCI is normalized using the degree day method. The HCIs and ELCIs are used to restore the yearly load profiles for heat and electricity purposes, respectively [5].

The ECI, HCI and ELCI for the different archetypes/building categories analyzed are shown in Table 2, all temperature-dependent heat consumption adjusted to the Trondheim normal climate using degree days.

Table 2

Average ECI, HCI and ELCI for the different archetypes/building categories analyzed adjusted to Trondheim normal climate using the degree days [4]

Archetype/building category	Specific energy consumption ($\text{kWh}/(\text{m}^2 \text{ year})$)		
	ECI	HCI	ELCI
Single family houses and apartment blocks ^a	166	116	49
Office buildings	235	100	135
Educational buildings: before 1997	175	109	69
Educational buildings: after 1997	174	103	69
Retirement homes (hospital buildings)	284	152	132
Hotels with restaurants	233	113	120

^a The total number does not add up with the numbers from HCI and ELCI due to different numbers and different buildings analyzed. The same applies for educational buildings because the ELCI for educational buildings was not divided into different archetypes.

The energy consumption indicators can incorporate changes in building design as well as introduction of new technology. HCIs and ELCIs for various archetypes/building categories should be calculated nationally to incorporate the various building design codes.

2.3.4. Coincidence factor

The sum of each customer's maximum load is not equal to the maximum load for the specified planning area, i.e. the maximum loads for all customers do not coincide [5]. This means that [10]:

$$(\Phi_1 + \Phi_2)_{\text{Maximum}} < \Phi_{1,\text{Maximum}} + \Phi_{2,\text{Maximum}} \quad (10)$$

Φ_1 , Φ_2 are the daily load for each building when design conditions occur.

The coincidence factor for N buildings has been defined by [10], among others:

$$\text{coincidence factor} = \frac{\Phi_{\text{Maximum (total)}}}{\sum_{n=1}^N \Phi_{n,\text{Maximum}}} \quad (11)$$

The generalized load profiles for the archetypes/building categories do not show the peak load demands for each building analyzed because the latter profiles include the coincidence factor due to the average expected value. The shapes of the various load profiles give the remaining coincidence factors for the heat and electricity loads [5].

2.3.5. Distribution losses

The distribution losses are strongly dependent upon the energy carriers, and electricity and district heating have different characteristics in relation to distribution losses. Maximum load losses and annual energy losses in the distribution systems are important parts of estimating heat and electricity demand for mixed energy distribution systems, and a brief overview is given in the paragraphs below.

2.3.5.1. Electricity. The losses in each level of the electricity grid, ΔP_i , may be written as shown below [9]:

$$\Delta P_i = \frac{R_i}{U_i^2} (P_i^2 + Q_i^2) \quad (12)$$

$$\Delta P_i = \frac{R_i}{U_i^2} (1 + \tan^2 \varphi_i) P_i^2 \quad (13)$$

$$\Delta P_i = k_i P_i^2 \quad (14)$$

where R_i is the resistance in grid level i in Ω ; U_i the voltage level on grid level i in kV; P_i the maximum local active power withdrawn on grid level i in kW; Q_i the appurtenant reactive power on grid level i in kvar; and $k_i = (R_i/U_i^2)(1 + \tan^2 \varphi_i)$.

The load losses in the grid are shown to be dependent on the load level at each time interval and this implies that the losses in the grid are dependent on season as well as day type and time of day. The annual electricity loss is the sum of the active load losses for every time interval throughout the year [5].

A simplified analysis of the electricity distribution losses has been performed in a case study, see Section 3.2. Distribution

losses from the grid company in Trondheim, Norway, have been collected. The maximum load loss is estimated to approximately 8% based on empirical data, whereas the annual electricity losses in the distribution grid varied between 5 and 6% in the most recent years based on real measurements [11].

2.3.5.2. District heating. The maximum heat loss and annual heat losses in a district heating distribution system are mainly dependent upon the following criteria [5]:

- High or low heat density within the planning area.
- Forward flow temperature and flow rates in the primary distribution system.
- Insulation standard and design of pipelines.
- Temperature efficiencies of the heat exchangers.

Each development project has to consider these criteria individually due to their complexity. However, new district heating distribution systems using single pipes systems may have an annual heat loss of approximately 10–15% and a load loss of about 2–3% at maximum heat load [12]. Twin pipes distribution systems will reduce the annual heat losses [13].

2.3.6. Algorithm for load aggregation

The procedure for the solution algorithm for aggregation of load profiles for a specified planning area with a given mixture of buildings is listed below [5] and in Fig. 6:

1. Select a specific planning area with a defined mixture of buildings.
2. Apply generalized heat and electricity load profiles for building based on the building category.
3. Use specific load indicators to construct real heat and electricity load profiles as well as standard deviations for design day.
4. Apply design reference year (DRY) [14] or other suitable yearly representations of the normal climate for calculating relative yearly load profiles. Use specific energy indicators to calculate real yearly heat and electricity load profiles.
5. Add real design heat and electricity load profiles at node connection points as well as standard deviations. Add yearly load profiles at the same node.
6. Add all design and yearly load profiles at the energy distribution/transformer unit, including a 95% quantile for peak load estimations.
7. Calculate coincidence factor for heat and electricity for design load profiles.
8. Choose energy carriers and include distribution losses for maximum load and annual energy accordingly.

3. Load profiles for heat and electricity demand in buildings

Resultant load profiles for various archetypes and building categories along with a case study will be presented in the sections below.

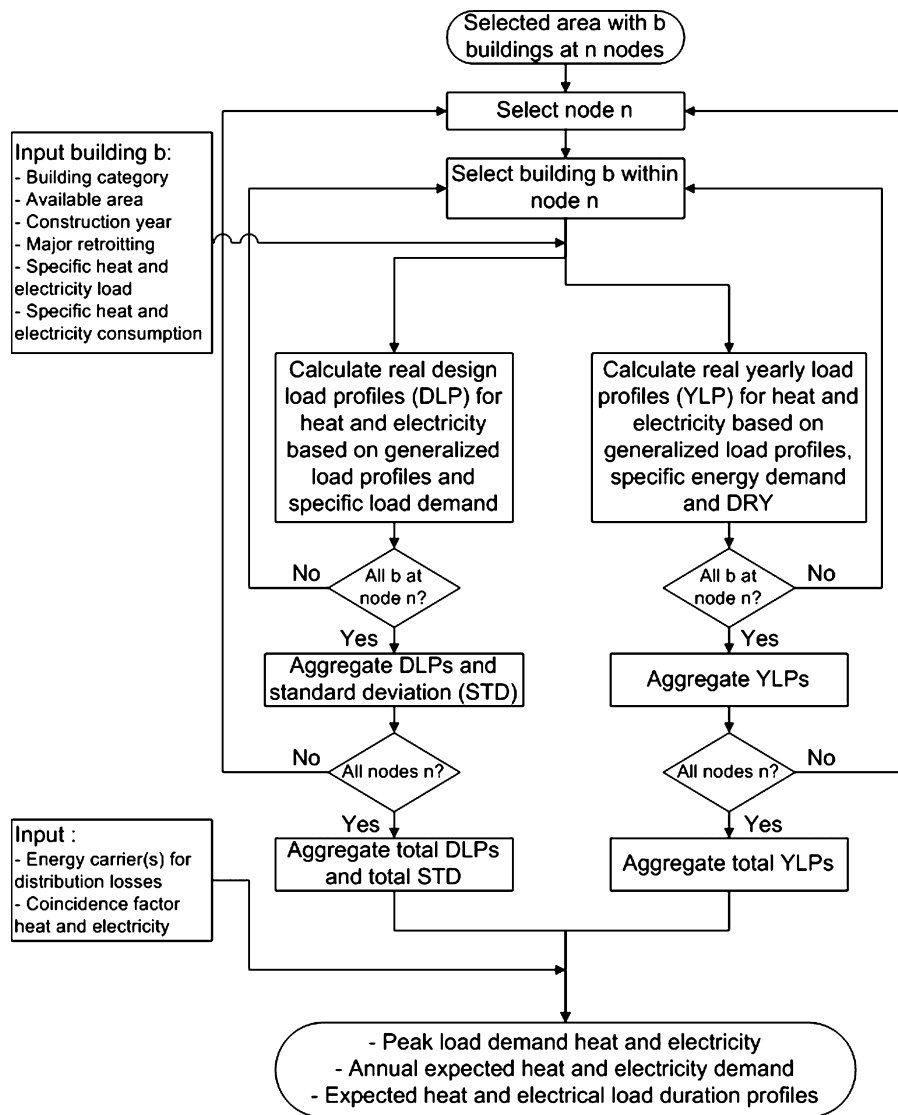


Fig. 6. Aggregation of generalized heat and electricity load profiles and energy demand for a specified planning area including distribution load and energy losses [4].

Generalized design heat load profiles AT1 WD and WE including STD EB

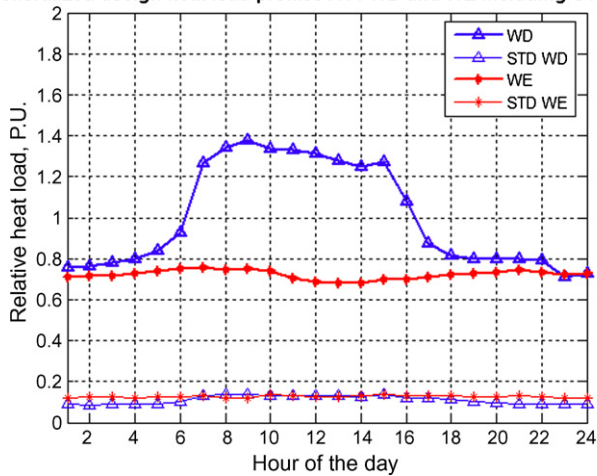


Fig. 7. Generalized design heat load profiles for educational buildings (EB), weekdays (WD) and weekends (WE), for archetype 1 (AT1); ventilation systems with time control for buildings built before 1997 [4].

Generalized design heat load profiles AT2 WD and WE including STD EB

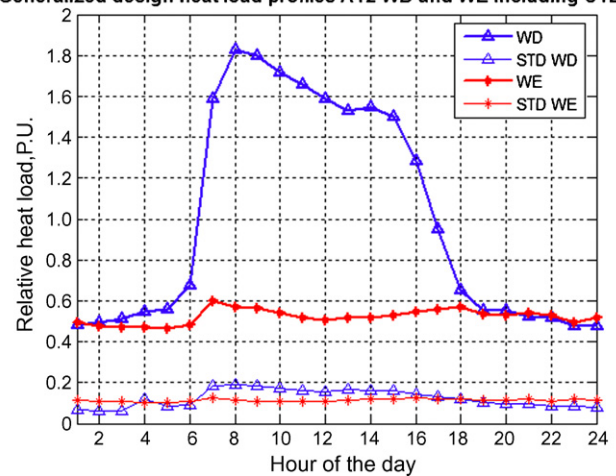


Fig. 8. Generalized design heat load profiles for educational buildings (EB), weekdays (WD) and weekends (WE), for archetype 2 (AT2); ventilation systems with time control for buildings built in 1997 and after [4].

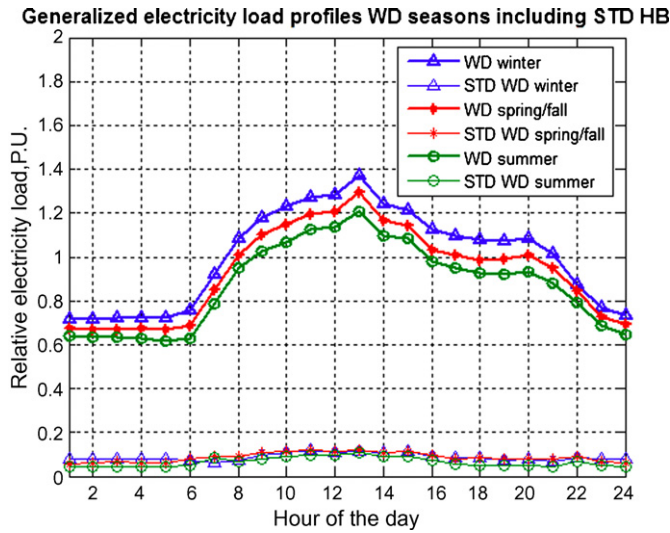


Fig. 9. Generalized electricity load profiles for hospital buildings for all seasons, weekdays, including standard deviation [4].

3.1. Load profiles for various archetypes/building categories

Load profiles are estimated for design conditions in Trondheim, Norway, i.e. design temperature of -19°C for heat load profiles and winter season for electricity load profiles. The building categories analyzed are single family houses and apartment blocks, office buildings, education buildings, retirement homes and hotels with restaurants.

3.1.1. Examples of generalized heat load profiles

The design heat load profiles are exemplified through the building category educational buildings. The design heat load profiles are presented by two archetypes (AT); buildings built before and after 1997 with ventilation systems running on time control. Changes in the building codes were given in 1997 in the Technical Regulations under the Norwegian Planning and

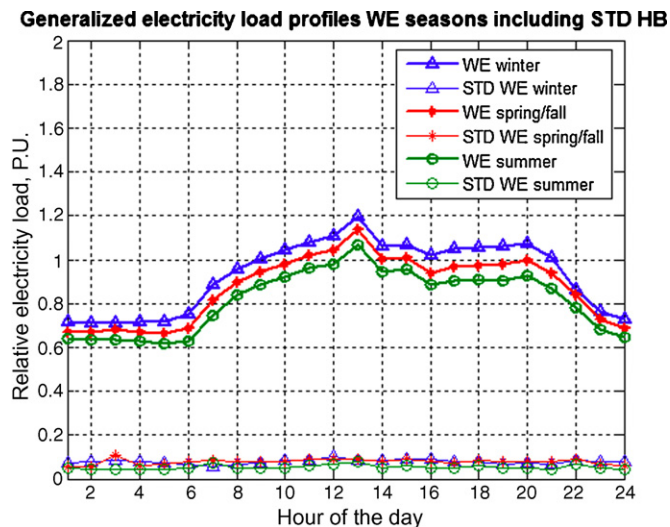


Fig. 10. Generalized electricity load profiles for hospital buildings for all seasons, weekends, including standard deviation [4].

Building Act [15]. More stringent requirements for the ventilation rate in new and retrofitted buildings was introduced, along with more stringent requirements for the coefficient of thermal transmittance for the building envelope. The generalized design heat load profiles for educational buildings archetype 1 and 2 including standard deviations (STD) for both day types, weekdays (WD) and weekends/holidays (WE), are shown in Figs. 7 and 8.

3.1.2. Examples of generalized electricity load profiles

The electricity load profiles are exemplified through the building category hospital buildings, i.e. this included only retirement homes. Figs. 9 and 10 show the electricity load profiles for all seasons including standard deviations (STD) for weekdays (WD) and weekends/holidays (WE), respectively. The peak loads for both day types occur at 1 p.m., but the load level during the weekdays is higher than during weekends for ordinary working hours. The fans operating the ventilation system ran on time control.

3.2. Case study

A case study is performed to show how the generalized load profiles can be applied to a specified planning area to estimate the design load profiles, yearly load profiles and annual energy demands, all divided into heat and electricity purposes. A description of the planning area, the solution procedure and the final results will be presented in the next sections.

3.2.1. Description of planning area

Several input variables are required to estimate the design load profile, yearly load profile and annual energy demand for a specified planning area. These are [5]:

1. Numbers of buildings within each archetype/building category.
2. Available area for each building.
3. Construction year for each building.
4. Major retrofitting, if any, for each building.
5. Type of heating system: hydronic heating system or electricity distribution system only.
6. Future development, if any, within the system boundaries.

The case study for this article is based on a fictitious development area located in Trondheim, Norway, climate. All the buildings within the system boundaries are defined to be

Table 3

Number of buildings located within the fictitious development area in Trondheim including average available area for every archetype/building category

Archetype/building category	Number	Average available area (m^2)
Single family houses	50	150
Apartment blocks	200	70
Office buildings	7	6000
Educational buildings archetype 2	3	2500
Nursing homes	5	3000
Hotels with restaurants	2	7000

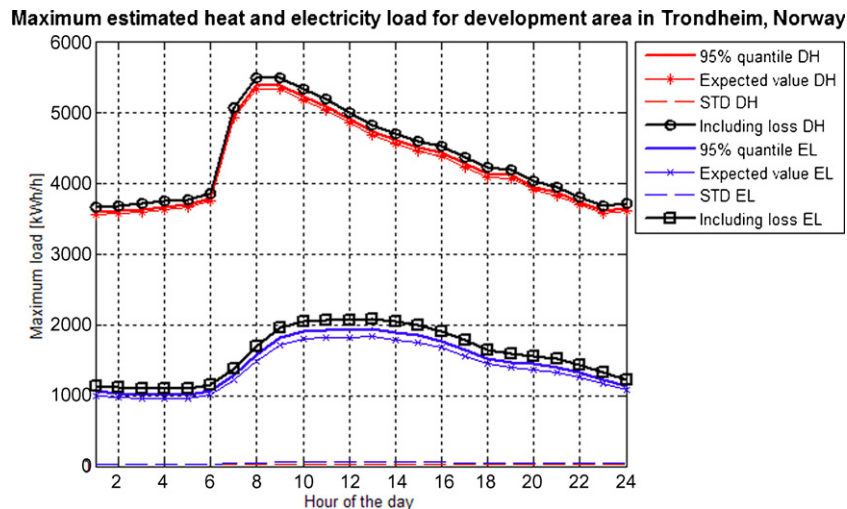


Fig. 11. Design load profiles for heat and electricity for development area in Trondheim, Norway.

built within the planning horizon and all buildings are assumed to have hydronic heating systems. Table 3 gives an overview of the buildings within the system boundaries along with the number of buildings and their available area.

To estimate the design heat load profile and the yearly heat load profile, the design temperature and outdoor temperatures from the design reference year (DRY) [14] are needed. The design temperature for Trondheim is -19°C . DRYs only exist for three locales in Norway, but not for Trondheim. As a consequence, the daily mean temperatures from the DRY for Oslo is applied due to a discrepancy of less than 2% between the normal climate in Trondheim and Oslo [16].

3.2.2. Solution procedure

Fig. 6 presented the solution procedure for the load aggregation method. First of all, the specified planning area has to be identified with all the required input parameters for the buildings within the system boundaries before the generalized load profiles can be applied. Second, the specific load indicators for all archetypes/building categories, as well as each buildings available area, are used to restore the design load profile for each building [5].

The generalized load profiles are also used to calculate the yearly load profiles and load duration profiles, all divided into heat and electricity. The HCIs and ELCIs, as well as available area, are applied to restore the generalized yearly load profiles for each building within the system boundaries [5].

The maximum load is estimated using the 95% t-quantile with $n - 1 = 266$ degrees of freedom, based on the number of buildings within the development area. The expected yearly load profiles for heat and electricity are estimated based on daily mean temperatures from DRY Oslo. The distribution losses are also included for both district heating and electricity supply.

3.2.3. Results

Fig. 11 shows the design load profiles for both heat and electricity demand including standard deviations, 95% quantiles and distribution losses at maximum load. The design or maximum loads for both purposes occur during weekdays.

The maximum heat load demand is estimated to 5.5 MW occurring at 8 a.m. during weekdays. A maximum load loss of 2% is included in the analysis for heat load design conditions. The heat load coincidence factor is calculated to 0.98 indicating homogeneous heat consumption patterns.

The maximum electricity load demand is estimated to 2.1 MW occurring at 1 p.m. during weekdays for the winter season. A maximum load loss of 8% is included in the analysis for electricity load design conditions. The electricity load coincidence factor is calculated to 0.94.

Fig. 12 shows the yearly and duration load profiles for heat and electricity for the development area in Trondheim, Norway. The annual distribution loss for district heating is set to 12%, whereas the annual distribution loss for the electricity grid is set to 5%. The annual energy demand for heating purposes is estimated to 12.7 GWh/year with a utilization time of 2300 h/year. The annual energy demand for electricity purposes is estimated to 11.4 GWh/year with a utilization time of 5400 h/year.

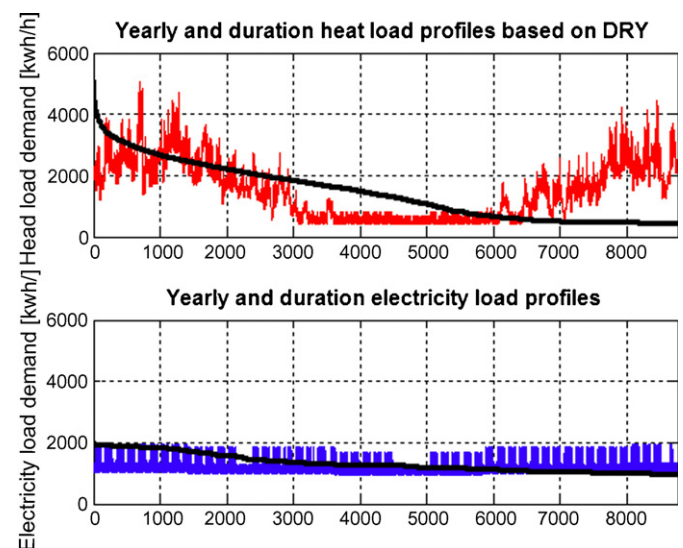


Fig. 12. Yearly and duration heat and electricity load profiles for development area in Trondheim, Norway.

The maximum heat load from the heat duration profile is slightly lower than the estimated design load profile. This is due to the use of two different indicators; specific heat load and specific heat consumption, as well as the lowest temperature in the DRY being 4 °C higher than the design temperature for Trondheim.

4. Conclusion

The load prediction method for heat and electricity demand in buildings presented in this article can be used for the purpose of planning for mixed energy distribution systems. The method is based on piece-wise regression analyses and probability distribution analyses of hourly measured district heat and electricity consumption in various buildings.

The load prediction method can estimate the design load profiles, yearly load profiles, duration load profiles and annual energy demands, all divided into heat and electricity purposes, within a specified planning area. The latter profiles are important input parameters to plan for the most economical, technical and environmental optimal energy distribution system for a planning area. The method can incorporate building categories such as single family houses and apartment blocks, office buildings, educational buildings, retirement homes and hotels with restaurants. Load profiles for other building categories can be developed when hourly measurements of heat and electricity are available.

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Linda Pedersen finished her master of science in Energy and Environmental Engineering in 2003 and her PhD degree in May 2007. The project that she was part of was called SEDS (sustainable energy distribution systems). The topic for her doctoral thesis was “Load modeling of buildings in mixed energy distribution systems”.

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